Project Overview – Kaan Olcar (S027334)

Project Title
Designing a Safe Landing

Project Goal
The aim of this project is to design a lightweight structure that can protect a fragile egg when dropped from a height. The structure should reduce the impact force during landing and prevent the egg from breaking.

Keywords

- Impact absorption
- Structural protection
- Suspended core
- Lightweight structure
- Shock distribution
- Controlled landing

Problem Understanding

Observation

Fragile objects such as eggs can easily break when dropped from a height because the impact force is directly transferred to the shell. Without a protective system, the egg receives the full force of the collision.

Problem Statement

How can a structure be designed to absorb impact forces and protect a fragile egg during a drop?

How Might We

How might we design a structure that controls impact and protects a fragile object during a fall?

Prompt:

Help me define the problem for an egg drop design project where the egg must survive a fall.

AI Response Summary

The AI helped clarify that the main challenge is managing the impact force during landing and separating the fragile object from direct contact with the ground.

Reflection

This helped me understand that the focus of the design should be impact control rather than only structural strength.

Ideation Phase

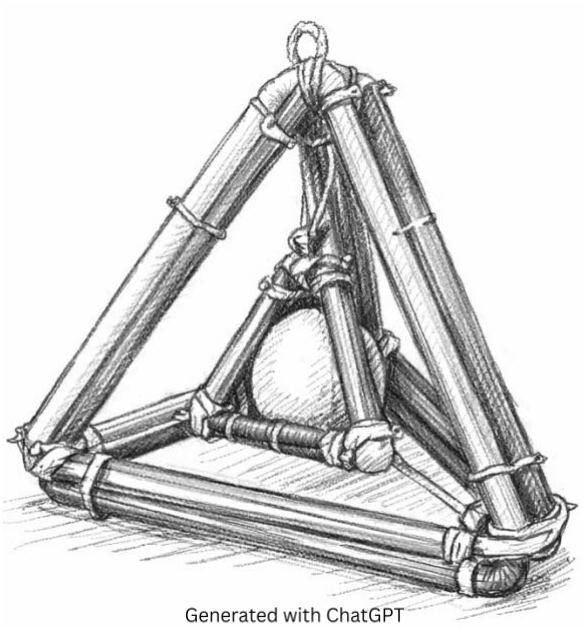
During the ideation stage, I explored different structural approaches that could reduce impact forces and protect the egg.

Prompt:

Suggest different design strategies for protecting an egg in an egg drop challenge.

AI Response Summary

The AI suggested several approaches such as cushioning materials, suspended structures, and protective frames that distribute impact forces.



Reflection

The suggestions helped me think about how the egg could be isolated from the main impact zone rather than simply surrounding it with protective material.

First Design Concept

Concept

The initial design used a triangular straw frame to create a protective structure around the egg.

The idea was that the frame would absorb part of the impact while the egg remained inside the structure.

Prompt:

Evaluate this triangular egg drop design and identify possible weaknesses.

AI Response Summary

The AI suggested that the outer frame might still transfer impact forces toward the center, which could still damage the egg.

Design Decision

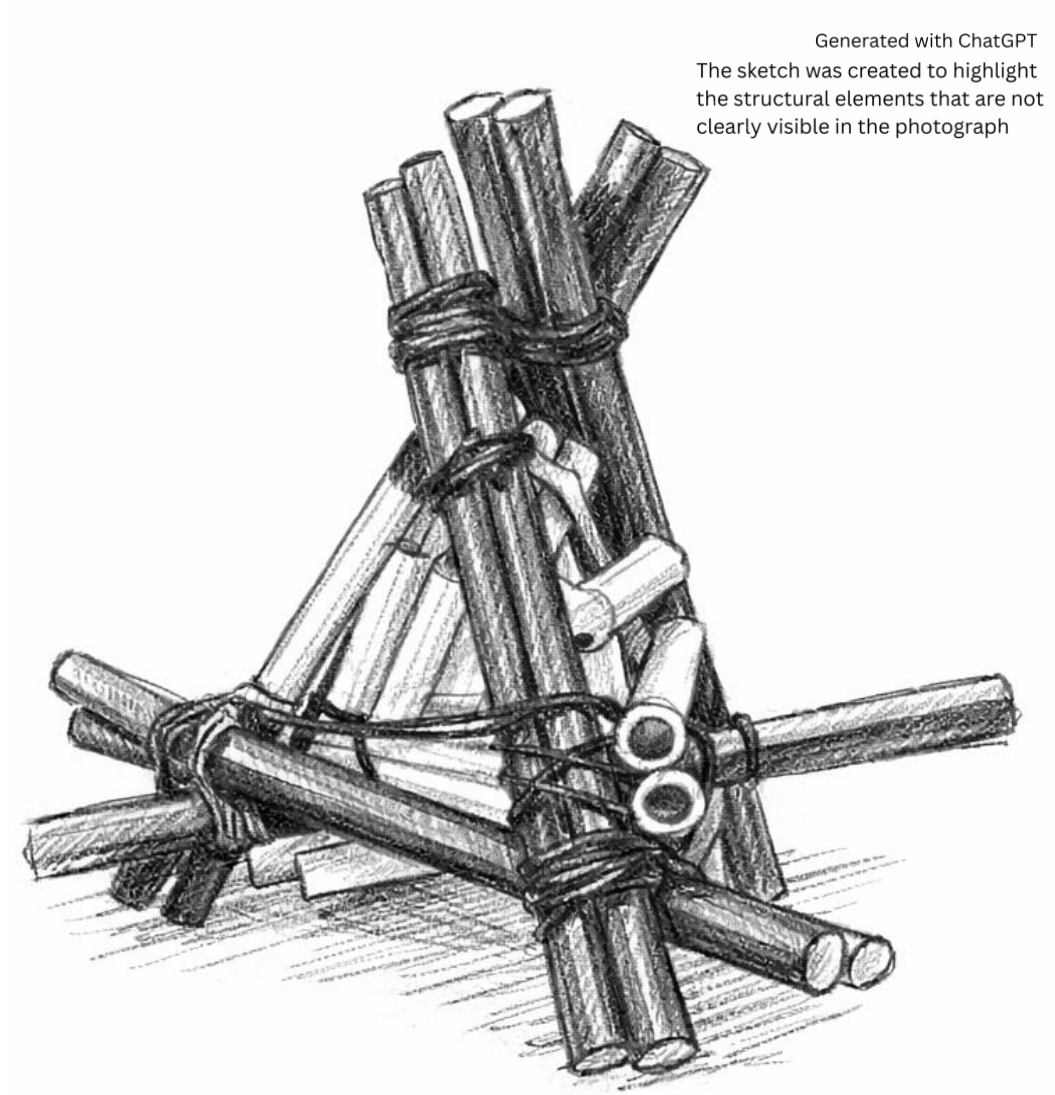
This feedback highlighted the need to better isolate the egg from the outer frame.

Reflection

This stage helped me understand that simply surrounding the egg with a structure is not enough. The design must separate the fragile object from the main impact path.

Design Development

To improve the design, I explored ways to better isolate the egg from the structure that receives the impact.



<u>Prompt:</u>

How can the egg be better isolated from impact forces in an egg drop structur

AI Response Summary

The AI suggested separating the impact frame from the protected core and using suspension principles to reduce the transfer of force.

Design Decisions

Based on this feedback, I redesigned the structure so that the egg would be suspended inside the outer frame rather than directly attached to it.

Accepted

- Suspended egg concept
- Impact absorbing outer frame
- Lightweight structure

Rejected

- Rigid designs where the egg touches the frame
- Structures that transfer impact directly to the center

Transformed

The triangular structure was modified to create a suspended inner core that protects the egg.

Final Design

Final Concept

The final design uses a triangular frame made from paper straws with the egg suspended inside the structure.

The outer frame absorbs and distributes the impact forces when the structure hits the ground. Because the egg is suspended inside the frame, it avoids direct contact with the ground and receives significantly less impact.

Prompt:

Review the improved suspended egg design and explain how it helps reduce impact.

AI Response Summary

The AI explained that separating the egg from the outer structure allows the frame to absorb impact energy before it reaches the egg.

Reflection

AI helped me analyze weaknesses in the first design and guided the transition toward a suspended protection concept.

Design Process Timeline

Problem Identification
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Idea Generation
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First Prototype (Triangular Frame)
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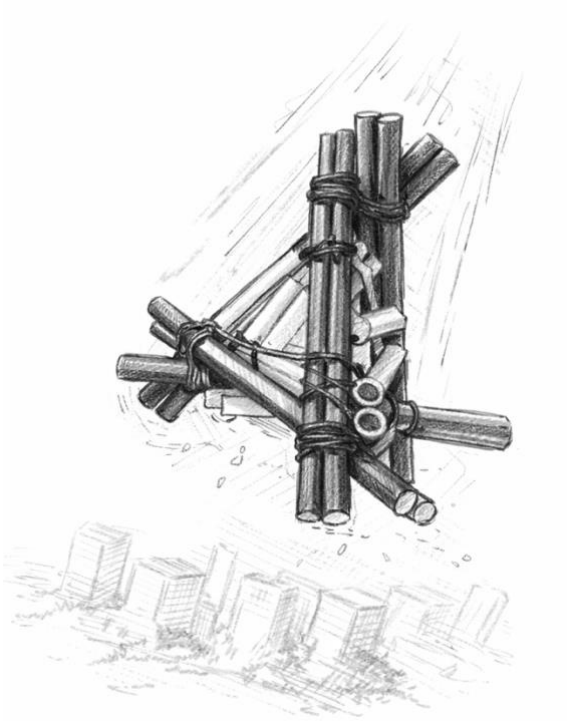
Evaluation and Critique
↓

Design Improvement (Suspended Egg Concept)
↓

Final Prototype



Final Reflection

AI supported my design process by helping me clarify the problem, generate ideas, and evaluate weaknesses in the initial concept. Through this iterative process, I improved the design by isolating the egg from the impact frame. This helped me develop a stronger and more effective final solution.